

Convexity adjustment for FRAs and futures with multicurves

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Abstract

The decoupling of discount and fixing curves changes the convexity adjustment for interest futures and induces convexity adjustment for forward rate agreements as they are normally traded. Both instruments therefore need to be convexity adjusted before a forward curve for swap pricing can be generated. In this paper we qualify the impact of this decoupling in the convexity adjustment and we derive the convexity adjustment formulas for futures and FRAs, given a HJM framework. We take into account that the discount curve is different from the forward curve and that they are partially decoupled. We are not aware of other studies which take the decoupling into account when calculating the convexity adjustments. Consequently this is the first time the convexity adjustment on forward rate agreements is reported, and this is the first time the convexity adjustment for futures is adjusted for decoupled discount rates.

1 Introduction

The decoupling of discount and fixing curves has been persistent since the credit crisis starting in 2007. The explanation of how credit wedge the OIS and the fixing curves apart is found in Fenger (2014). Here we will study how this decoupling changes the convexity adjustments for futures, and induces convexity adjustment for FRAs, the traded version of forward rate agreements.

By construction then the forward rates needed to price a swap can be read directly of the relevant forward curve. However, the cash flow in FRAs and futures is discounted differently and consequently adjustments are needed to convert a forward rate from a forward curve into a FRA or future quote. Since the differences in discounting imply convexity differences between a single floating period in a swap and a FRA and a future, we use the term convexity adjustment.

Define Gamma as the second order derivative with respect to the relevant forward rate then a future, to a good approximation, does not have any Gamma. A swaplet, receiving one future fixing and paying a fixed amount, have negative Gamma due to the discounting by the risk free rate. Hence a convexity adjustment is subtracted from the futures quote to imply the corresponding swaplet rate. A FRA is discounted harder than the swaplet, as the discounting applied is the risk free rate until fixing and the discounting applied in the forward period is the expected fixing, which is higher than the risk free rate and positively correlated with the risk free rate. In section 6 we find that the correlation is indeed positive. Therefore a FRA has an even lower Gamma than the swaplet, and hence a convexity adjustment must be added to the FRA quote to imply the corresponding swaplet rate.

The convexity adjustment observed in the market, ie the convexity adjustment implied directly from market prices can differ significantly from the convexity adjustment implied from a dynamic model. This difference does not necessarily invalidate the model. Rather the market convexity adjustment is resulting from the market flow, ie supply/demand. An extreme case happened in the spring of 2013 where FRAs traded higher than futures. At the 21st of May 2013 the EUR future contract expiring in two years were bid 1.2bps below the corresponding EUR FRA bid, see Carver (2013), where the traditionally one-curve model predicted a negligible spread of opposite sign. The observed spread could modelwise be interpreted as an expectation of an extreme

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negative correlation between LIBOR and discount rates and high expected volatility. A more likely explanation is the rise of central counterparty clearing houses where swaps and FRAs are cleared, while futures are traded on exchanges. As margin is required at both places, then banks have an incentive to not have large opposite positions. Traditionally the swap desks are not charged for excessive usage of margin, but as more and more banks becomes aware of this mismatch in incentives, then the drift for netting out positions will continue.

We base our calculations of the convexity adjustment on discretely compounded rates, as opposed to continuously compounded rates, as the underlying market rate are discretely compounded and since difference between the two methods can be in the range of the bid-offer spread of affected market instruments, see Fenger (2015).

In this contribution we focus exclusively on the model induced convexity adjustment. Our starting point is the HJM framework. The rest of the paper is organized as follows. Section 4 introduces the model capturing the dynamics between the disc and LIBOR curve in the case of instantaneous rates. Section 5 introduces the model capturing the dynamics in the case of money market rates. Expressions for the convexity adjustment for futures and FRAs are calculated. Section 6 states empiric observations, and give numeric estimates for the convexity adjustments. Section 7 concludes. Useful definitions and results are given in appendix A.

2 Intuitive explanation

The convexity adjustment for futures turns out to be positive:

$$\text{Implied futures rate} = \text{FRA rate} + \text{Convexity adjustment} \quad (1)$$

The intuitive argument for this is that a long (BPV positive) futures contract, with daily margin payments, is better than a long FRA contract, with payment at expiry. The reason for this is that in case of an interest rate rise then the gain from the FRA contract is relatively diminished by a higher discount factor, whereas in the case of an interest rate fall then the loss is relatively enlarged. This implies that the FRA contract has negative Gamma, whereas the futures contract has zero Gamma, and buyers of the contracts should therefore be compensated by a lower FRA rate. Note that collateralization of the FRA contract does not change the issue, when the yield on collateral resembles to the risk free rate, which is the typical case.

3 Historical overview

Flesaker (1993) calculates the convexity adjustment in the Ho-Lee model for futures based on discretely compounded rate. Further he considers the effect of the marked-to-market frequency. In Kirikos and Novak (1997) the convexity adjustment is found for the Hull-White model for futures on discretely compounded rates and thus a generalization of Flesaker (1993).

Here we shall assume continuous marked-to-market payments of futures, and disregard initial and maintenance margin.

4 Model embracing credit spread - instantaneous rates

From Duffie and Singleton (1997) we have (ignoring the loss rate for simplicity) that

$$\tilde{f}(t, T) = f(t, T) + h(t, T) \quad (2)$$

As noted in section A.5.1 then f is a martingale under Q^T . Carrying through the same arguments we note that also $\tilde{f}$ is a martingale. As consequence of (2) then also h is a martingale under Q^T , and therefore dh is a pure diffusion process.

Using that $\tilde{f}$ is a martingale we define $\sigma_{\tilde{f}}(t, T)$:

$$d\tilde{f}(t, T) = \sigma_{\tilde{f}}(t, T)dW^{Q^T}(t) \quad (3)$$

When the correlation between $d\tilde{f}$ and dh is $\rho_{\tilde{f},h}(t, T)$ and introducing the volatility $\sigma_h(t, T)$:

$$dh(t, T) = \sigma_h(t, T) \left(\frac{\rho_{\tilde{f},h}(t, T)}{\sqrt{1 - \rho_{\tilde{f},h}^2(t, T)}} \right) \cdot \begin{pmatrix} dW^Q(t) \\ dW_{\perp}^Q(t) \end{pmatrix} \quad (4)$$

where W^{Q^T} and $W_{\perp}^{Q^T}$ are independent.

The dynamics of the risk free rate can be expressed as the dynamics of the risky forward rate and some independent noise:

$$df(t, T) \stackrel{(2)(3)(4)}{=} \begin{pmatrix} \sigma_{\tilde{f}}(t, T) - \rho_{\tilde{f},h}(t, T)\sigma_h(t, T) \\ -\sigma_h(t, T)\sqrt{1 - \rho_{\tilde{f},h}^2(t, T)} \end{pmatrix} \cdot \begin{pmatrix} dW^{Q^T}(t) \\ dW_{\perp}^{Q^T}(t) \end{pmatrix} \quad (5)$$

For later use we note that df can be expressed as $d\tilde{f}$ and a part independent of $d\tilde{f}$:

$$df(t, T) = \beta^{f\tilde{f}}(t, T)d\tilde{f}(t, T) + \epsilon_{\perp\tilde{f}}(t, T) \quad (6)$$

where

$$\begin{aligned} \beta^{f\tilde{f}}(t_1, t_2) &\equiv 1 - \frac{\rho_{\tilde{f},h}(t_1, t_2)\sigma_h(t_1, t_2)}{\sigma_{\tilde{f}}(t_1, t_2)} \\ \epsilon_{\perp\tilde{f}}(t_1, t_2) &\equiv -\sigma_h(t_1, t_2)\sqrt{1 - \rho_{\tilde{f},h}^2(t_1, t_2)} \end{aligned}$$

We can interpret $\beta^{f\tilde{f}}$ as the linear regression coefficient from df explained by $d\tilde{f}$.

4.1 Convexity adjustment

To find the futures convexity adjustment, we need to find the difference between the futures quote and the forward rate. From (55) we see that the futures quote is given by the Q -expectation such that the convexity adjustment becomes:

$$\text{CA}_{\text{FUT}}(t, T; \tilde{f}) = \mathbb{E}_t^Q [\tilde{f}(T, T)] - \tilde{f}(t, T)$$

To calculate the Q -expectation, we need to determine $d\tilde{f}$ under Q . $d\tilde{f}$ is known under Q^T so a measure change will be performed by applying the Girsanov theorem with the Girsanov kernel. The Girsanov kernel is found by lemma A.5. To apply this lemma we need to know the p -dynamics under Q in terms of W and $W_{\perp}$.

Under Q the p -dynamics will have this structure:

$$dp(t, T) = r(t)p(t, T)dt + \sigma_p(t, T)p(t, T) \begin{pmatrix} dW^Q(t) \\ dW_{\perp}^Q(t) \end{pmatrix} \quad (7)$$

where $\sigma_p(t, T)$ is found in a similar fashion to the method used for deriving the volatility in (60):

$$\sigma_p(t, T) \stackrel{(5)}{=} - \int_t^T \begin{pmatrix} \sigma_{\tilde{f}}(t, s) - \rho_{\tilde{f},h}(t, s)\sigma_h(t, s) \\ \sigma_h(t, s)\sqrt{1 - \rho_{\tilde{f},h}^2(t, s)} \end{pmatrix} ds \quad (8)$$

According to lemma A.5 we see that the Girsanov kernel is given by (8) such that (58) translates into

$$\begin{pmatrix} dW^{Q^T}(t) \\ dW_{\perp}^{Q^T}(t) \end{pmatrix} \rightarrow \begin{pmatrix} dW^Q(t) \\ dW_{\perp}^Q(t) \end{pmatrix} + \sigma_p(t, T)dt \quad (9)$$

From (3) (8) and (9) we get the dynamics of $\tilde{f}$ under Q :

$$d\tilde{f}(t, T) = \sigma_{\tilde{f}}(t, T) \left(\int_t^T \sigma_{\tilde{f}}(t, s) - \rho_{\tilde{f},h}(t, s)\sigma_h(t, s) ds \right) dt + \sigma_{\tilde{f}}(t, T)dW^Q(t)$$

which is the risky equivalent of (66), the HJM drift condition. Whereby the convexity adjustment becomes

$$\text{CA}_{\text{FUT}}(t, T; \tilde{f}) = \mathbb{E}_t^Q \left[\int_t^T \sigma_{\tilde{f}}(u, T) \left(\int_u^T \sigma_{\tilde{f}}(u, s) \beta^{f\tilde{f}}(u, s) ds \right) du \right] \quad (10)$$

For constant $\sigma_{\tilde{f}}$ and $\beta^{f\tilde{f}}$, ie the Ho-Lee model generalized to a multicurve setup, we get

$$\text{CA}_{\text{FUT}}(t, T; \tilde{f}) = \frac{1}{2} \beta^{f\tilde{f}} \sigma_{\tilde{f}}^2 (T - t)^2$$

The above result gives us a hint on what to expect from more elaborate calculations. In the risk free world then $\sigma_h = 0$ and then $\beta^{f\tilde{f}} = 1$ and the above equation reduces to the classical results for convexity adjustment for instantaneous rates. If the risk free rates were zero and non-rates were due to credit only then $\beta^{f\tilde{f}} = 0$ and the convexity adjustment would be zero. It seems fair to expect that in the real world then $0 < \beta^{f\tilde{f}} < 1$, and that the convexity adjustment implied from multicurve models therefore will be reduced roughly by the fraction $\beta^{f\tilde{f}}$ compared to the implied convexity adjustment from singlecurve models.

5 Model embracing credit spread - money market rates

We will find the dynamics of the risky forward rates under Q . Then by calculating the expectation and subtracting the forward rate we achieve the futures convexity adjustment. According to Duffie and Singleton (1997) then for rates from T_1 to T_2 :

$$\tilde{F}(t, T_1, T_2) = F(t, T_1, T_2) + H(t, T_1, T_2) \quad (11)$$

$\tilde{F}(t, T_1, T_2)$ and $F(t, T_1, T_2)$ are simple rates and H is the implied credit spread. Section A.1.4 and A.1.5 state that F and $\tilde{F}$ are Q^{T_2} martingales. Hence H is also a martingale.

Let the Pearson correlation between the daily changes of F and H be denoted by $\rho_{\tilde{F}, H}$. This correlation and (11) then implies

$$dH(t, T_1, T_2) = \sigma_H(T_1, T_2) \rho_{\tilde{F}, H}(T_1, T_2) dW^{Q^{T_2}}(t) + \sigma_H(T_1, T_2) \sqrt{1 - \rho_{\tilde{F}, H}^2(T_1, T_2)} dW_{\perp}^{Q^{T_2}}(t) \quad (12)$$

$$dF(t, T_1, T_2) = (\sigma_{\tilde{F}}(T_1, T_2) - \sigma_H(T_1, T_2) \rho_{\tilde{F}, H}(T_1, T_2)) dW^{Q^{T_2}} - \sigma_H(T_1, T_2) \sqrt{1 - \rho_{\tilde{F}, H}^2(T_1, T_2)} dW_{\perp}^{Q^{T_2}} \quad (13)$$

$$d\tilde{F}(t, T_1, T_2) = \sigma_{\tilde{F}}(T_1, T_2) dW^{Q^{T_2}}(t) \quad (14)$$

Here we have introduced the credit spread volatility σ_H . (13) and (14) imply that dF can be expressed as $d\tilde{F}$ and some independent noise:

$$dF(t, T_1, T_2) = \beta^{F\tilde{F}}(T_1, T_2) \cdot d\tilde{F}(t, T_1, T_2) + \epsilon_{\perp \tilde{F}}(T_1, T_2) \quad (15)$$

where

$$\begin{aligned} \beta^{F\tilde{F}}(T_1, T_2) &\equiv 1 - \frac{\rho_{\tilde{F}, H}(T_1, T_2) \sigma_H(T_1, T_2)}{\sigma_{\tilde{F}}(T_1, T_2)} \\ \epsilon_{\perp \tilde{F}}(T_1, T_2) &\equiv \sigma_H(T_1, T_2) \sqrt{1 - \rho_{\tilde{F}, H}^2(T_1, T_2)} \end{aligned} \quad (16)$$

An important observation here is that the independence of $d\tilde{F}$ and $\epsilon_{\perp \tilde{F}}$ and the assumption that $\epsilon_{\perp \tilde{F}}$ is normally distributed, makes $\beta^{F\tilde{F}}(T_1, T_2)$ the regression coefficient in linear regression. As indicated by (10) in the instantaneous case, $\beta^{F\tilde{F}}(T_1, T_2)$ plays a key role for determining the convexity adjustment for traded futures. Theoretically $\beta^{F\tilde{F}}(T_1, T_2)$ could be implied from current market data. However, this estimate will be uncertain due to market imperfections and temporary flow of market orders.

We get the dynamics of H under Q :

$$dH(t, T_1, T_2) \stackrel{(8)(9)(12)}{=} -\sigma_H(T_1, T_2) \left(\frac{\rho_{\tilde{F}, H}(T_1, T_2)}{\sqrt{1 - \rho_{\tilde{F}, H}^2(T_1, T_2)}} \right) \sigma_p(t, T_2) dt + \sigma_H(T_1, T_2) \left(\frac{\rho_{\tilde{F}, H}(T_1, T_2)}{\sqrt{1 - \rho_{\tilde{F}, H}^2(T_1, T_2)}} \right) \cdot \left(dW_{\perp}^{Q^T} \right) \quad (17)$$

With

$$F(t, T_1, T_2) = \frac{1}{\delta_{12}} \left(\frac{p(t, T_1)}{p(t, T_2)} - 1 \right) \Rightarrow dF = d \left(\frac{p(t, T_1)}{p(t, T_2)} \right) \quad (18)$$

then we get using Itô's lemma

$$\begin{aligned} dF(t, T_1, T_2) &\stackrel{(64)}{=} \frac{1}{\delta_{12}} \frac{1}{p(t, T_2)} dp(t, T_1) - \frac{1}{\delta_{12}} \frac{p(t, T_1)}{p^2(t, T_2)} dp(t, T_2) \\ &\quad + \frac{1}{\delta_{12}} \frac{p(t, T_1)}{p^3(t, T_2)} d^2 p(t, T_2) - \frac{1}{p^2(t, T_2)} \frac{1}{\delta_{12}} dp(t, T_1) dp(t, T_2) \end{aligned} \quad (19)$$

Applying (7) to (19) we get dF under Q :

$$dF = \frac{1}{\delta_{12}} \frac{p(t, T_1)}{p(t, T_2)} \sigma_p(t, T_2) \sigma_p(T_1, T_2) dt - \frac{1}{\delta_{12}} \frac{p(t, T_1)}{p(t, T_2)} \sigma_p(T_1, T_2) \cdot \left(dW_{\perp}^{Q^T} \right) \quad (20)$$

Equating the volatilities in (13) and (20) it is evident that $\sigma_p(T_1, T_2)$ can be formulated with the simple rates parameters:

$$\sigma_p(T_1, T_2) = -\delta_{12} \frac{p(t, T_2)}{p(t, T_1)} \begin{pmatrix} \sigma_{\tilde{F}}(T_1, T_2) - \sigma_H(T_1, T_2) \rho_{\tilde{F}, H}(T_1, T_2) \\ -\sigma_H(T_1, T_2) \sqrt{1 - \rho_{\tilde{F}, H}^2(T_1, T_2)} \end{pmatrix} \quad (21)$$

Trivial use of (8)(11)(17)(20) and (21) yields $d\tilde{F}$ under Q :

$$d\tilde{F} = \sigma_{\tilde{F}}(T_1, T_2) \int_t^{T_2} \sigma_{\tilde{f}}(t, s) - \rho_{\tilde{f}, h}(t, s) \sigma_h(t, s) ds dt + \sigma_{\tilde{F}}(T_1, T_2) dW^Q \quad (22)$$

5.1 Convexity adjustment for futures - in general

Inserting (8) into (22):

$$d\tilde{F} = -\sigma_{\tilde{F}}(T_1, T_2) \sigma_{p1}(t, T_2) dt + \sigma_{\tilde{F}}(T_1, T_2) dW^Q \quad (23)$$

Divide time into equidistant parts:

$$t_0 < t_1 < t_2 < \dots < t_i < \dots < t_{N-1} < t_N$$

where $t_{N-1} = T_1$ and $t_N = T_2$. Then using (8)

$$\sigma_{p1}(t, T_2) = \sum_{i=0}^{N-1} \sigma_{p1}(t_i, t_{i+1}) \quad (24)$$

Inserting (24) into (23)

$$d\tilde{F}(t, T_1, T_2) = -\sigma_{\tilde{F}}(T_1, T_2) \sum_{i=0}^{N-1} \sigma_{p1}(t_i \vee t, t_{i+1}) \mathbf{1}_{t < t_{i+1}} dt + \sigma_{\tilde{F}}(T_1, T_2) dW^Q(t) \quad (25)$$

where the maximum operator and the indicator function comes into play as time passes forward. Inserting a generalization of (21) into (25)

$$\begin{aligned} d\tilde{F}(t, T_1, T_2) &= \sigma_{\tilde{F}}(T_1, T_2) \sum_{i=0}^{N-1} \delta_{i, i+1} \left(\sigma_{\tilde{F}}(t_i \vee t, t_{i+1}) - \sigma_H(t_i \vee t, t_{i+1}) \rho_{\tilde{F}, H}(t_i \vee t, t_{i+1}) \right) \mathbf{1}_{t < t_{i+1}} \frac{p(t, t_{i+1})}{p(t, t_i)} dt \\ &\quad + \sigma_{\tilde{F}}(T_1, T_2) dW^Q(t) \end{aligned} \quad (26)$$

Integrating (26) from t_0 to T_1 :

$$\begin{aligned}\tilde{F}(T_1, T_1, T_2) &= \sigma_{\tilde{F}}(T_1, T_2) \sum_{i=0}^{N-1} \delta_{i,i+1} \int_{t_0}^{T_1} \left(\sigma_{\tilde{F}}(t_i \vee s, t_{i+1}) - \sigma_H(t_i \vee s, t_{i+1}) \rho_{\tilde{F},H}(t_i \vee s, t_{i+1}) \right) 1_{s < t_{i+1}} \frac{p(s, t_{i+1})}{p(s, t_i)} ds \\ &\quad + \sigma_{\tilde{F}}(T_1, T_2) (W^Q(T_1) - W^Q(t_0)) + \tilde{F}(t_0, T_1, T_2) \\ &\stackrel{1}{\approx} \sigma_{\tilde{F}}(T_1, T_2) \sum_{i=0}^{N-1} \delta_{i,i+1} \sigma_{\tilde{F}}(t_i, t_{i+1}) \beta^{F\tilde{F}}(t_i, t_{i+1}) \int_{t_0}^{t_{i+1}} \frac{p(s, t_{i+1})}{p(s, t_i)} ds \\ &\quad + \sigma_{\tilde{F}}(T_1, T_2) (W^Q(T_1) - W^Q(t_0)) + \tilde{F}(t_0, T_1, T_2)\end{aligned}\tag{27}$$

With

$$\beta^{F\tilde{F}}(t_i, t_{i+1}) = 1 - \frac{\rho_{\tilde{F},H}(t_i, t_{i+1}) \sigma_H(t_i, t_{i+1})}{\sigma_{\tilde{F}}(t_i, t_{i+1})}$$

The inequality is caused by the σ values are sometimes being overvalued by the removal of the maximum operator. Taking expectation of (27) and subtracting $\tilde{F}(t_0, T_1, T_2)$:

$$\text{CA}_{\text{FUT}}(t_0) \approx \sigma_{\tilde{F}}(T_1, T_2) \sum_{i=0}^{N-1} \delta_{i,i+1} \sigma_{\tilde{F}}(t_i, t_{i+1}) \beta^{F\tilde{F}}(t_i, t_{i+1}) \mathbb{E}_t^Q \left[\int_{t_0}^{t_{i+1}} \frac{p(s, t_{i+1})}{p(s, t_i)} ds \right]$$

And using another approximation² our main equation for the futures convexity adjustment becomes:

$$\boxed{\text{CA}_{\text{FUT}}(t_0) \approx \sigma_{\tilde{F}}(T_1, T_2) \sum_{i=0}^{N-1} \delta_{i,i+1} \sigma_{\tilde{F}}(t_i, t_{i+1}) (t_{i+1} - t_0) \beta^{F\tilde{F}}(t_i, t_{i+1})}\tag{28}$$

Note that the effect of the decoupling $\beta^{F\tilde{F}}(t_i, t_{i+1}) \neq 1$ on the convexity adjustment, is that a weighted $\beta^{F\tilde{F}}$ is multiplied to the standard one curve. The advantage of (28) is that it only depends on observable market inputs.

With constant parameters we get that (28) simplifies to a familiar expression times the regression coefficient³:

$$\text{CA}_{\text{FUT}}(t_0) = \frac{1}{2} \sigma_{\tilde{F}}^2(T_1, T_2) (T_2 - t_0) (T_2 + (T_2 - T_1) - t_0) \beta^{F\tilde{F}}(T_1, T_2)$$

5.2 Convexity adjustment for futures - risk free case

We can get the convexity adjustment as a special case of the risky case. Set $\beta^{F\tilde{F}} = 1$, $\sigma_h = \sigma_H = 0$. (22) then reduces to:

$$dF(t, T_1, T_2) = \int_t^{T_2} \sigma_f(t, s) ds \sigma_F(t, T_1, T_2) dt + \sigma_F(t, T_1, T_2) dW^Q(t)\tag{29}$$

¹In most terms the maximum operator does not have any effect since $s \leq t_i$. In the remaining terms we apply:

$$\sigma_{\tilde{F}}(t_i \vee s, t_{i+1}) - \sigma_H(t_i \vee s, t_{i+1}) \rho_{\tilde{F},H}(t_i \vee s, t_{i+1}) \approx \sigma_{\tilde{F}}(t_i, t_{i+1}) - \sigma_H(t_i, t_{i+1}) \rho_{\tilde{F},H}(t_i, t_{i+1})$$

² $\mathbb{E}_t^Q \left[\frac{p(s, t_{i+1})}{p(s, t_i)} \right] \lesssim 1$ for a normal yield curve and $\gtrsim 1$ for an inverse yield curve.

³And using $\delta_{i,i+1} = t_{i+1} - t_i$ and using

$$\sum_{i=0}^{N-1} \delta_{i,i+1} (t_{i+1} - t_0) = \Delta t \sum_{i=0}^{N-1} 1 = \frac{1}{2} \Delta t N \cdot \Delta t (N+1) = \frac{1}{2} (T_2 - t_0) \cdot (T_2 + (T_2 - T_1) - t_0)$$

The result could be compared with the Ho-Lee result for futures based on continuously compounded rates:

$$\text{CA}_{\text{FUT}}(t_0) = \frac{1}{2} \sigma_{\tilde{F}}^2(T_1, T_2) (T_1 - t_0) (T_2 - t_0)$$

(21) implies:

$$\sigma_F(t, T_1, T_2) = \frac{1}{\delta_{12}} \frac{p(t, T_1)}{p(t, T_2)} \int_{T_1}^{T_2} \sigma_f(t, s) ds \quad (30)$$

Inserting (30) into (29):

$$dF = \int_{T_1}^{T_2} \sigma_f(t, s) ds \int_t^{T_2} \sigma_f(t, s) ds \left(\frac{1}{\delta_{12}} \frac{p(t, T_1)}{p(t, T_2)} \right) dt + \int_{T_1}^{T_2} \sigma_f(t, s) ds \left(\frac{1}{\delta_{12}} \frac{p(t, T_1)}{p(t, T_2)} \right) dW^Q \quad (31)$$

From (18) then the expectation of (31) at T_1 is given by the Geometric Brownian Motion formula:

$$\mathbb{E}_t^Q \left[\frac{1}{\delta_{12}} \frac{p(T_1, T_1)}{p(T_1, T_2)} \right] = \exp(S(t, T_1, T_2) \int_t^{T_2} S(t, s, T_2) ds) \quad (32)$$

where

$$S(t, T_1, T_2) = \int_{T_1}^{T_2} \sigma_f(t, s) ds \quad (33)$$

Inserting (30) into (33) then the expression in in market observable parameters:

$$S(t, T_1, T_2) = \delta_{12} \frac{p(t, T_2)}{p(t, T_1)} \sigma_F(t, T_1, T_2) \quad (34)$$

Ie (56) (18) and (32) yields

$$\text{CA}_{\text{FUT}}(t) = \left(\exp \left\{ S(t, T_1, T_2) \int_t^{T_1} S(t, s, T_2) ds \right\} - 1 \right) \left(\frac{1}{\delta_{12}} + F(t, T_1, T_2) \right) \quad (35)$$

This result (35) is similar with the result for risk free futures given in Piterbarg and Renedo (2006).

5.3 Convexity adjustment for futures

- independence between credit spread and risk free rates

In this section we assume independence of the credit spread and the risk free rates under the risk neutral measure, and the T_2 forward measure. Among others this implies that dH and dF are independent and H is independent of the risk free cash account, which is defined like:

$$B(t, T) = \exp\left(\int_t^T r(s) ds\right)$$

From Hunt and Kennedy (2000) the convexity adjustment is given by:

$$\text{CA}_{\text{FUT}}(t) = \frac{\text{cov}_t^Q(\tilde{F}, B(t, T))}{p(t, T_2)} \quad (36)$$

From (11) and (36) and using $\text{cov}_t^Q(H, B(t, T)) = 0$:

$$\text{CA}_{\text{FUT}}(t) = \frac{\text{cov}_t^Q(F, B(t, T))}{p(t, T_2)} \quad (37)$$

This implies that any convexity adjustment formula can be used as long as one applies the parameters for the risk free rate F . We now show what this implies in the framework here. We can use the independence between dH and dF to eliminate the parameter σ_H :

$$\begin{aligned} & \left(\frac{\sigma_H(T_1, T_2) \rho_{\tilde{F}, H}(T_1, T_2)}{\sigma_H(T_1, T_2) \sqrt{1 - \rho_{\tilde{F}, H}^2(T_1, T_2)}} \right) \cdot \left(\frac{\sigma_{\tilde{F}}(T_1, T_2) - \sigma_H(T_1, T_2) \rho_{\tilde{F}, H}}{-\sigma_H(T_1, T_2) \sqrt{1 - \rho_{\tilde{F}, H}^2(T_1, T_2)}} \right) = 0 \\ & \Rightarrow \sigma_H(T_1, T_2) = \rho_{\tilde{F}, H}(T_1, T_2) \sigma_{\tilde{F}}(T_1, T_2) \end{aligned} \quad (38)$$

With (13) and (38):

$$dF(t, T_1, T_2) = \sigma_{\tilde{F}}(T_1, T_2) \sqrt{1 - \rho_{\tilde{F}, H}^2(T_1, T_2)} dW_F^{Q^{T_2}} \quad (39)$$

where $W_F^{Q^{T_2}}$ is Brownian motion and defined as

$$dW_F^{Q^{T_2}} \equiv \sqrt{1 - \rho_{\tilde{F}, H}^2(T_1, T_2)} dW^{Q^{T_2}} - \rho_{\tilde{F}, H}(T_1, T_2) dW_{\perp}^{Q^{T_2}}$$

From (39) and (37) then it is seen that we can apply formula (35) with the volatility $\sigma_F(t, T_1, T_2)$ in (34) replaced with $\sigma_{\tilde{F}}(t, T_1, T_2) \sqrt{1 - \rho_{\tilde{F}, H}^2(T_1, T_2)}$.

5.4 Convexity adjustment for FRAs

In this section we carry through the calculations using non-constant parameters. The standard traded forward rate agreement, a FRA, pays

$$(\tilde{F}(T_1, T_1, T_2) - K) \delta_{12} / (1 + \tilde{F}(T_1, T_1, T_2) \delta_{12})$$

at T_1 , where K is the strike. The present value is given by:

$$\text{PV}(t) = p(t, T_1) \mathbb{E}_t^{Q^{T_1}} \left[\frac{(\tilde{F}(T_1, T_1, T_2) - K) \delta_{12}}{1 + \tilde{F}(T_1, T_1, T_2) \delta_{12}} \right] \quad (40)$$

Setting the present value to zero allows us to solve for K :

$$K = \frac{\mathbb{E}_t^{Q^{T_1}} \left[\frac{\tilde{F}(T_1, T_1, T_2)}{1 + \delta_{12} \tilde{F}(T_1, T_1, T_2)} \right]}{\mathbb{E}_t^{Q^{T_1}} \left[\frac{1}{1 + \delta_{12} \tilde{F}(T_1, T_1, T_2)} \right]} \quad (41)$$

If the convexity adjustment were zero then K would be the forward rate, but as argued in the introduction we expect the convexity adjustment to be negative. Below we shall seek a first order approximation to the convexity adjustment. Applying the expansion $1/(1+x) \approx 1-x$ numerous times to (41):

$$K \approx \mathbb{E}_t^{Q^{T_1}} [\tilde{F}(T_1, T_1, T_2)] - \delta_{12} \left(\mathbb{E}_t^{Q^{T_1}} [\tilde{F}^2(T_1, T_1, T_2)] - \mathbb{E}_t^{Q^{T_1}} [\tilde{F}(T_1, T_1, T_2)]^2 \right)$$

Subtracting the forward rate from the equation above we can write down the convexity adjustment:

$$\text{CA}_{\text{FRA}}(t) \approx \left(\mathbb{E}_t^{Q^{T_1}} [\tilde{F}(T_1, T_1, T_2)] - \mathbb{E}_t^{Q^{T_2}} [\tilde{F}(T_1, T_1, T_2)] \right) - \delta_{12} \left(\mathbb{E}_t^{Q^{T_1}} [\tilde{F}^2(T_1, T_1, T_2)] - \mathbb{E}_t^{Q^{T_1}} [\tilde{F}(T_1, T_1, T_2)]^2 \right) \quad (42)$$

Ie that convexity adjustment stems from two term. The first is the effect of a change between two forward measures. The second is the volatility squared times the coverage. Applying (9) we can change the measure from Q^{T_2} to Q^{T_1} :

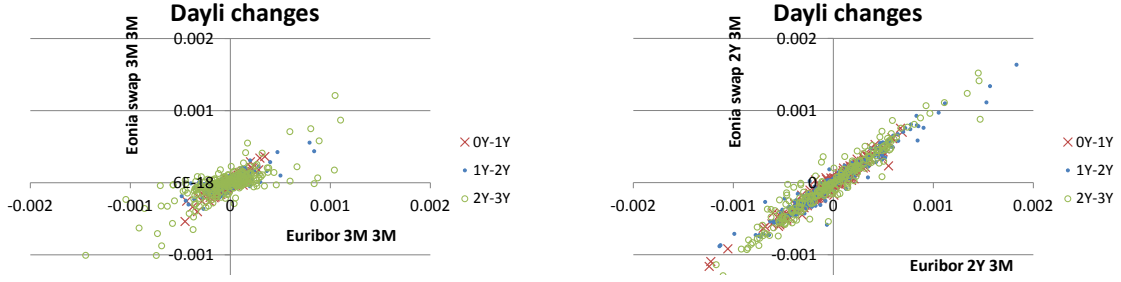
$$dW^{Q^{T_2}} \rightarrow dW^{Q^{T_1}} + \sigma_p(T_1, T_2) dt \quad (43)$$

With (43) and $\sigma_p(T_1, T_2)$ given by (21) we get $d\tilde{F}$ under Q^{T_1} :

$$d\tilde{F} = \sigma_{\tilde{F}}(t, T_1) \delta_{12} \frac{p(t, T_1)}{p(t, T_2)} \left(\sigma_{\tilde{F}}(t, T_1) - \sigma_H(t, T_1) \rho_{\tilde{F}, H}(t, T_1) \right) dt + \sigma_{\tilde{F}}(t, T_1) dW^{Q^{T_1}} \quad (44)$$

Since the drift of $\tilde{F}(T_1, T_1, T_2)$ is zero under Q^{T_2} then we can replace the first part of (42) with the integrated drift from (44):

$$\mathbb{E}_t^{Q^{T_1}} [\tilde{F}(T_1, T_1, T_2)] - \mathbb{E}_t^{Q^{T_2}} [\tilde{F}(T_1, T_1, T_2)] = \sigma_{\tilde{F}}(t, T_1, T_2) \delta_{12} \frac{p(t, T_1)}{p(t, T_2)} \left(\sigma_{\tilde{F}}(t, T_1) - \sigma_H(t, T_1) \rho_{\tilde{F}, H}(t, T_1) \right) (T_1 - t) \quad (45)$$



(a) Daily changes of the forward discount rate versus the risky forward starting in 3 months. Linear regression coefficient for entire period 0Y-3Y: $dF = \beta^{F\tilde{F}} d\tilde{F} + \epsilon \Rightarrow \beta^{F\tilde{F}} = 0.567 \pm 0.017$. The correlation between $d\tilde{F}$ and dH is 0.69. The differentials denote daily changes. $\text{std.dev}(\text{residuals}) = 6.1\text{e-}5$.

(b) Daily changes of the forward discount rate versus the risky forward starting in 2 years. Linear regression coefficient for entire period 0Y-3Y: $dF = \beta^{F\tilde{F}} d\tilde{F} + \epsilon \Rightarrow \beta^{F\tilde{F}} = 0.920 \pm 0.008$. The correlation between $d\tilde{F}$ and dH is 0.34. The differentials denote daily changes. $\text{std.dev}(\text{residuals}) = 7.1\text{e-}5$.

Figure 1: Comparing the 3 month forward rate on the discount curve (Eonia) with the 3 month forward rate on the Libor curve (Euribor). Figures (a) and (c) display the absolute rate level for forward rates starting in 2 years and 3 month respectively. Figures (b) and (d) display the corresponding daily changes. Data from 23Mar2012-24Mar2015.

In the second term of (42) we have the volatility of $\tilde{F}(T_1, T_1, T_2)$:

$$\mathbb{E}_t^{Q^{T_1}} [\tilde{F}^2(T_1, T_1, T_2)] - \mathbb{E}_t^{Q^{T_1}} [\tilde{F}(T_1, T_1, T_2)]^2 = \sigma_{\tilde{F}}^2(t, T_1, T_2)(T_1 - t) \quad (46)$$

Adding (45) and $\delta_{12} \cdot (46)$ together yields the convexity adjustment to first order for a FRA:

$$\text{CA}_{\text{FRA}}(t) \approx -\delta_{12} \sigma_{\tilde{F}}^2(t, T_1, T_2)(1 - \beta^{F\tilde{F}}(t, T_1))(T_1 - t) \quad (47)$$

In (47) we have used $p(t, T_1)/p(t, T_2) \approx 1$. In the current low volatility environment the adjustment is negligible, but in a high volatility environment it can be of the order of a basis point. Note that the convexity adjustment is negative, cf. the discussion in the introduction. It seems most practical to imply $\beta^{F\tilde{F}}$ from historical data as implied from (15) and imply $\sigma_{\tilde{F}}$ from current market data.

6 Empiric observations

In figure 1 we plot daily changes of the forward rates representing expected 3 months fixings versus forward starting OIS swaps of the same duration both EUR rates. The latter is a de facto discount rate instrument. In figure 1a and 1b the start tenor is 3 months and 2 years, respectively. Figure 2 summarizes similar plots for different tenors and different currencies. It is evident that the decoupling is strongest for short tenors and particular strong for USD. The USD and GBP models are cross currency adjusted to EUR. However, the numbers do not change significantly if silo models are used, ie models where the domestic OIS rates act as discount rates. The $\beta^{F\tilde{F}}$ coefficients were about the same for 6 months forward rates.

7 Conclusion

We derive expressions for the convexity adjustment of interest futures and FRAs fixing and discount curves are decoupled. We find that decoupling lowers the convexity adjustment for futures and induces convexity adjustment for FRAs. With historical estimates of the coupling of between discount and fixing curves, in a low rate / low volatility environment like EUR, we find that the convexity adjustment of futures is slightly lower than traditional estimates, and that the convexity adjustment for forward rate agreements is below the size of the bid-offer spread.

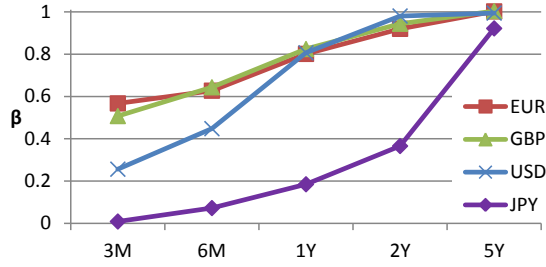


Figure 2: Linear regression coefficient $\hat{\beta}^{F\tilde{F}}(0, T)$ such that $dF = \hat{\beta}^{F\tilde{F}}(0, T) d\tilde{F} + \epsilon$ as function of start tenor of forward rates. Observations based on data from 23Mar2012-24Mar2015 for 3month forward rates.

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Appendix

A Definitions and classic results

A.1 Risk free rates

Let the risk free rates at time t be implied from the set of zero coupon bonds $\{p(t, T)\}_T$.

A.1.1 Instantaneous forward rate

The instantaneous forward rates on the discount curve is defined from

$$p(t, T) = \exp\left(-\int_t^T f(t, s)ds\right)$$

It then follows

$$f(t, T) = -\frac{1}{p(t, T)} \frac{\delta p(t, T)}{\delta T}$$

Alternatively define $f(t, T)$ as the instantaneous rate at time T set at time t . $f(t, T)$ can then be derived from the replication argument of buying $p(t, T + \Delta T)/p(t, T)$ number of $p(t, T)$ and selling one $p(t, T + \Delta T)$. The price at time t of this portfolio is 0. At time T we receive $p(t, T + \Delta T)/p(t, T)$ and at time $T + \Delta T$ we pay one. Taking $\Delta T \rightarrow 0$ this implies a forward rate of

$$f(t, T) = -\frac{1}{p(t, T)} \frac{\delta p(t, T)}{\delta T} \quad (48)$$

A.1.2 Bank account

Define the bank account as

$$B(t, T) = \exp\left(\int_t^T r(s) ds\right)$$

where r is the short rate and $r(t) = f(t, t)$. From risk neutral evaluation

$$p(t, T) = \mathbb{E}_t^Q[1/B(t, T)]$$

A.1.3 The forward rate is the expected short rate

At time t then $f(T, T)$ is a stochastic variable. The price at time t of the pay off $f(T, T) - K$ is given by

$$\begin{aligned} \text{PV} &= \mathbb{E}_t^Q\left[\frac{1}{B(t, T)}(f(T, T) - K)\right] \\ &= p(t, T)(\mathbb{E}_t^{Q^T}[f(T, T)] - K) \\ \text{PV} &= 0 \Rightarrow K = \mathbb{E}_t^{Q^T}[f(T, T)] \end{aligned}$$

I.e. at price=0 we can at time t borrow at time T at a rate equal to $\mathbb{E}_t^{Q^T}[f(T, T)]$. Cf. section A.1.1 this rate was defined as $f(t, T)$. I.e.

$$\begin{aligned} f(t, T) &= \mathbb{E}_t^{Q^T}[f(T, T)] \\ &= \mathbb{E}_t^{Q^T}[r(T)] \end{aligned} \quad (49)$$

Alternatively note from (48) that $-\frac{\delta p(t, T)}{\delta T}$ is a martingale under Q^T and that

$$\lim_{t \rightarrow T} -\frac{\delta p(t, T)}{\delta T} = f(T, T) = r(T)$$

I.e.

$$f(t, T) = \mathbb{E}_t^{Q^T}[r(T)]$$

A.1.4 Simple risk free forward rate

By a replication argument the simple risk free forward rate is determined by tradable zero coupon bonds:

$$F(t, T_1, T_2) = \left(\frac{p(t, T_1)}{p(t, T_2)} - 1\right)/\delta_{12} \quad (50)$$

I.e. $F(t, T_1, T_2)$ is a martingale under Q^{T_2} and in particular:

$$F(t, T_1, T_2) = \mathbb{E}_t^{Q^{T_2}}[F(T_1, T_1, T_2)] \quad (51)$$

A.1.5 Simple risky forward rate

Let $\tilde{F}(t, T_1, T_2)$ be the simple forward rate on a defaultable loan. A forward rate agreement with strike K payable at time T_2 then has the value:

$$PV = p(t, T_2)(\mathbb{E}_t^{Q^{T_2}}[\tilde{F}(T_1, T_1, T_2)] - K) \quad (52)$$

A value of zero implies:

$$K = \mathbb{E}_t^{Q^{T_2}}[\tilde{F}(T_1, T_1, T_2)] \quad (53)$$

which implies the risky forward rate to be K . Ie the risky forward rate is a martingale and in particular

$$\tilde{F}(t, T_1, T_2) = \mathbb{E}_t^{Q^{T_2}}[\tilde{F}(T_1, T_1, T_2)] \quad (54)$$

A.2 Futures

Lemma A.1. *Let $\Phi(t)$ be any process. Let T be given. If $\Phi(t) = \mathbb{E}_t[\Phi(T)] \forall t \leq T$ then for any s such that $t \leq s \leq T$:*

$$\Phi(t) = \mathbb{E}_t[\Phi(s)]$$

Proof: By rule of iterated expectations:

$$\begin{aligned} \Phi(t) &= \mathbb{E}_t[\Phi(T)] \\ &= \mathbb{E}_t[\mathbb{E}_s[\Phi(T)]] \\ &= \mathbb{E}_t[\Phi(s)] \end{aligned}$$

◦

Note If $\Phi(t) = \mathbb{E}_t[\Phi(T)]$ for all t and T such that $t \leq T$ then $\Phi(t)$ is a martingale.

Proposition A.2. *Let $\Phi(t)$ be a futures process such that $\Phi(t) = \Psi(T)$ at $t = T$. Then $\Phi(t)$ is a Q -martingale, and*

$$\Phi(t) = \mathbb{E}_t^Q[\Psi(T)] \quad (55)$$

Note $\Psi(T)$ is some stochastic variable known at time T . Here it is relevant to think of $\Psi(T)$ as the Libor fixing fixed at time T , $\tilde{F}(T, T, T + \Delta T)$.

Proof: (Arguments copied from Andersen and Piterbarg (2010). An earlier reference is Hunt and Kennedy (2000).) Since the value of a futures contract always is zero then

$$\mathbb{E}_t^Q\left[\int_t^\tau \frac{1}{B(t, s)} d\Phi(s)\right] = 0$$

where

$$dB(t, s) = B(t, s)r(s)ds \text{ and } B(t, t) = 1$$

This holds for all t and τ such that $t \leq \tau \leq T$ and since $B(t, s)$ bounded and positive then

$$\mathbb{E}_t^Q[d\Phi(s)] = 0 \forall s$$

and therefore $\Phi(t)$ is a martingale. The expectation follows directly from the fact that $\Phi(t)$ is a martingale. ◦

A.3 Convexity Adjustment Definition

We define the convexity adjustment as the difference between the market quote K of either FRAs or futures, and the forward fixing at T_1 for the forward rate between T_1 and T_2 :

$$CA(t, T_1) = K(t, T_1) - \tilde{F}(t, T_1, T_2) \quad (56)$$

where $\tilde{F}(t, T_1, T_2)$ is the risky (contains a credit element) forward rates.

A.4 Girsanov

Theorem A.3. Girsanov's. *If Q_1 and Q_2 are equivalent measures then there exists a ϕ (the Girsanov kernel) such that*

$$dW^{Q_1} = \phi(t)dt + dW^{Q_2} \quad (57)$$

Proof. See Bjork (2009), prop 11.3. $\square$

Proposition A.4. *Assume numeraire S_i for Q_i and $dS_i = \dots dt + \sigma_i(t)S_i(t)dW^{Q_i}$ where σ_i is adapted to W^{Q_i} for $i \in \{1, 2\}$. Then the Girsanov kernel for the transition from Q_1 to Q_2 as defined by (57) is given by*

$$\phi(t) = \sigma_2(t) - \sigma_1(t)$$

Proof. See Bjork (2009), prop 26.5. $\square$

Corollary A.5. Going from Q to Q^T *From theorem A.3 and prop. A.4 with $S_1(t) = B(t)$ and $S_2(t) = p(t, T)$ with the dynamics $dp(t, T) = \dots dt + \sigma_p(t, T)p(t, T)dW$ and $dB(t) = r(t)B(t)dt$ then*

$$dW^Q = \sigma_p(t, T)dt + dW^{Q^T} \quad (58)$$

A.5 Convexity adjustment in the instantaneous case

A.5.1 f Dynamics under Q^T

From (49) and lemma A.1 it is evident that $f(\cdot, T)$ is a martingale under Q^T and

$$df(t, T) = \sigma_f(t, T)dW^{Q^T}(t) \quad (59)$$

where $\sigma_f(t, T)$ is adapted to W^{Q^T} .

A.5.2 p Dynamics under Q

Under Q we know that the drift of the bond is $r(t)$:

$$dp(t, T) = r(t)p(t, T)dt + \sigma_p(t, T)p(t, T)dW^Q \quad (60)$$

$$\text{where} \quad \sigma_p(t, T) \equiv - \int_t^T \sigma_f(t, s)ds \quad (61)$$

$\sigma_p(t, T)$ is found as follows. From Ito and (60):

$$d \log p(t, T) = \dots dt + \sigma_p(t, T)dW$$

Since $\log p(t, T) = - \int_t^T f(t, u)du$ then integrate both sides of (59) wrt T and compare:

$$\begin{aligned} d \log p(t, T) &= d\left(- \int_t^T f(t, u)du\right) \\ &\stackrel{\forall t_0}{=} d(f(t_0, u) + \int_t^T \int_{t_0}^t \sigma_f(s, u) dW^{Q^T} du) \\ &= d(f(t_0, u) + \int_{t_0}^t \int_t^T \sigma_f(s, u) du dW^{Q^T}) \\ &= \int_t^T \sigma_f(s, u) du dW^{Q^T} \end{aligned}$$

A.5.3 p Dynamics under Q^T

Doing a Girsanov transformation by inserting (58) in (60) we get the p -dynamics under Q^T :

$$dp(t, T) = (r(t) + \sigma_p^2(t, T))p(t, T)dt + \sigma_p(t, T)p(t, T)dW^{Q^T} \quad (62)$$

A.5.4 p_1/p_2 dynamics

Define

$$S(t, T_1, T_2) \equiv - \int_{T_1}^{T_2} \sigma_f(t, s) ds \quad (63)$$

From Itô's lemma:

$$d \frac{p(t, T_1)}{p(t, T_2)} = \frac{1}{p(t, T_2)} dp(t, T_1) - \frac{p(t, T_1)}{p^2(t, T_2)} dp(t, T_2) + \frac{p(t, T_1)}{p^3(t, T_2)} d^2 p(t, T_2) - \frac{1}{p^2(t, T_2)} dp(t, T_1) dp(t, T_2) \quad (64)$$

Under Q^Z , $Z \geq t$ ($Q^t = Q$):

$$d \frac{p(t, T_1)}{p(t, T_2)} = S(t, Z, T_2) S(t, T_1, T_2) \frac{p(t, T_1)}{p(t, T_2)} dt - S(t, T_1, T_2) \frac{p(t, T_1)}{p(t, T_2)} dW^{Q^Z}$$

The expectation at $T \leq T_1$ and $T \leq Z$:

$$\mathbb{E}_t^{Q^Z} \left[\frac{p(T, T_1)}{p(T, T_2)} \right] = \frac{p(t, T_1)}{p(t, T_2)} \exp(S(t, Z, T_2) S(t, T_1, T_2) \cdot (T - t))$$

The expectation at $T \leq T_1$ and $Z = t$:

$$\mathbb{E}_t^{Q^Z} \left[\frac{p(T, T_1)}{p(T, T_2)} \right] = \frac{p(t, T_1)}{p(t, T_2)} \exp(S(t, T_1, T_2) \int_t^T S(t, s, T_2) ds) \quad (65)$$

In particular under Q^{T_1} :

$$d \frac{p(t, T_1)}{p(t, T_2)} = S^2(t, T_1, T_2) \frac{p(t, T_1)}{p(t, T_2)} dt - S(t, T_1, T_2) \frac{p(t, T_1)}{p(t, T_2)} dW^{Q^{T_1}}$$

The expectation at $T \leq T_1$:

$$\mathbb{E}_t^{Q^{T_1}} \left[\frac{p(T, T_1)}{p(T, T_2)} \right] = \frac{p(t, T_1)}{p(t, T_2)} \exp(S^2(t, T_1, T_2) \cdot (T - t))$$

In case that $T = T_1$:

$$\mathbb{E}_t^{Q^{T_1}} \left[\frac{1}{p(T_1, T_2)} \right] = \frac{p(t, T_1)}{p(t, T_2)} \exp(S^2(t, T_1, T_2) \cdot (T_1 - t))$$

A.5.5 f Dynamics under Q

From Girsanov (58):

$$dW^{Q^T} = -\sigma_p(t, T) dt + dW^Q$$

Inserting this into (59) and applying (61):

$$df(t, T) = \sigma_f(t, T) \int_t^T \sigma_f(t, s) ds + \sigma_f(t, T) dW^Q(t) \quad (66)$$

(66) is the main result in Heath et al. from 1992.

A.5.6 Convexity adjustment for a future on the instantaneous risk free rate

Integrating (66):

$$\begin{aligned} f(T, T) &= f(t, T) + \int_t^T \sigma_f(u, T) \int_u^T \sigma_f(u, s) ds du + \int_t^T \sigma_f(t, T) dW^Q \\ \Rightarrow E_t^Q[f(T, T)] &= f(t, T) + \mathbb{E}_t^Q \left[\int_t^T \sigma_f(u, T) \int_u^T \sigma_f(u, s) ds du \right] \end{aligned} \quad (67)$$

I.e. the convexity adjustment for a future on the instantaneous risk free rate becomes

$$\text{CA}(t, T; f) = \mathbb{E}_t^Q \left[\int_t^T \sigma_f(u, T) \int_u^T \sigma_f(u, s) ds du \right] \quad (68)$$

If $\sigma_f(t, T) = \sigma_f$ is constant, ie the Ho-Lee model, then

$$\text{CA}(t, T; f) = \frac{1}{2} \sigma_f^2 (T - t)^2$$